

National University of Sciences and Technology

Department of Mechatronics Engineering

EE-313: Electronic Circuit Design

Complex Engineering Problem (CEP) Report

**Noise Mitigation and Multi-Stage
Signal Recovery System**

Submitted By: Muhammad Tahir Hammad
Nouman Nadeem Cheema

Instructor: Dr. Zaki Uddin

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1 Problem Definition

In radio frequency (RF) communication and audio signal processing, maintaining signal integrity under noisy conditions is a fundamental engineering challenge. Practical signal sources particularly low power broadcast receivers deliver signals contaminated with broadband **white noise**, characterized by erratic voltage spikes and rapid fluctuations uniformly distributed across the frequency spectrum.

This project addresses that challenge by engineering a complete hardware system capable of:

1. Receiving a noisy, low-amplitude input signal.
2. Suppressing broadband high frequency noise through cascaded passive filtering.
3. Amplifying the recovered signal through a multi-stage BJT amplifier.
4. Driving a standardized speaker load of $R_L = 5\text{ k}\Omega$ with a stable, noise-free output.

Success is defined by achieving a measurable improvement in Signal-to-Noise Ratio (SNR) at the load, while maintaining waveform fidelity and sufficient voltage gain.

2 Technical Context & Literature Review

2.1 Signal-to-Noise Ratio (SNR)

SNR quantifies the ratio of desired signal power to background noise power, expressed in decibels:

$$\text{SNR}_{\text{dB}} = 10 \log_{10} \left(\frac{P_{\text{signal}}}{P_{\text{noise}}} \right) = 20 \log_{10} \left(\frac{V_{\text{signal,rms}}}{V_{\text{noise,rms}}} \right) \quad (1)$$

White noise is particularly challenging because its energy is spread uniformly across all frequencies, requiring precise band-limiting to preserve the desired signal while rejecting noise.

2.2 Passive RC Filtering—Second-Order Networks

A first-order RC low-pass filter provides a roll-off of -20 dB/decade beyond its cut-off frequency. For aggressive noise rejection, a second-order (cascaded dual-stage) RC network is employed, achieving -40 dB/decade roll-off. The transfer function of a second-order passive RC filter is:

$$H(s) = \frac{1}{(1 + sR_1C_1)(1 + sR_2C_2)} \quad (2)$$

The cut-off frequency for each identical stage ($R_1 = R_2 = R$, $C_1 = C_2 = C$) is:

$$f_c = \frac{1}{2\pi RC} \quad (3)$$

The chosen topology resembles a Butterworth type approximation in the sense that the passband is maximally flat, avoiding amplitude ripple within the audio band.

2.3 BJT Amplifier Configurations

2.3.1 Common Emitter (CE) Stage Voltage Gain

The CE configuration provides high voltage gain with a 180° phase inversion. The small signal voltage gain is:

$$A_{v1} = -\frac{R_C \parallel R'_L}{r_e} = -\frac{\beta(R_C \parallel R'_L)}{r_\pi} \quad (4)$$

where $r_e = V_T/I_C \approx 26 \text{ mV}/I_C$ at room temperature, R_C is the collector resistor, and R'_L is the effective AC load seen at the collector.

2.3.2 Common Collector (CC) Stage Impedance Buffer

The CC (emitter-follower) stage provides unity voltage gain ($A_v \approx 1$) with a very high input impedance and low output impedance, making it ideal as an output buffer to drive the $5 \text{ k}\Omega$ load without loading the CE gain stage:

$$A_{v2} = \frac{R_E}{R_E + r_e} \approx 1 \quad (5)$$

2.4 Miller Effect and Miller Compensation

In a CE amplifier, feedback capacitance C_f between collector and base is “Miller multiplied” at the input:

$$C_{\text{Miller}} = C_f(1 - A_{v1}) \quad (6)$$

With $A_{v1} \approx -254 \text{ V/V}$, even a small $C_f = 100 \text{ pF}$ creates a large reflected capacitance at the input, forming a dominant pole that limits the upper -3 dB frequency. This is the mechanism of **Miller compensation**: by deliberately introducing $C_4 = 100 \text{ pF}$ from collector to base, the high-frequency bandwidth is controlled, preventing oscillation and ensuring stability.

3 Circuit Design and Methodology

3.1 System Architecture Overview

The implemented system consists of three cascaded functional blocks, as shown in Figure ??:

1. **Block 1 Dual-Stage RC Low-Pass Filter:** Suppresses high-frequency noise above $f_c \approx 7.23 \text{ kHz}$.
2. **Block 2 CE Amplifier (Q1, BC547):** Provides the primary voltage gain of $\approx -254 \text{ V/V}$ with Miller compensation via $C_4 = 100 \text{ pF}$.
3. **Block 3 CC Buffer (Q2, BC547):** Matches impedance to the $5 \text{ k}\Omega$ speaker load.

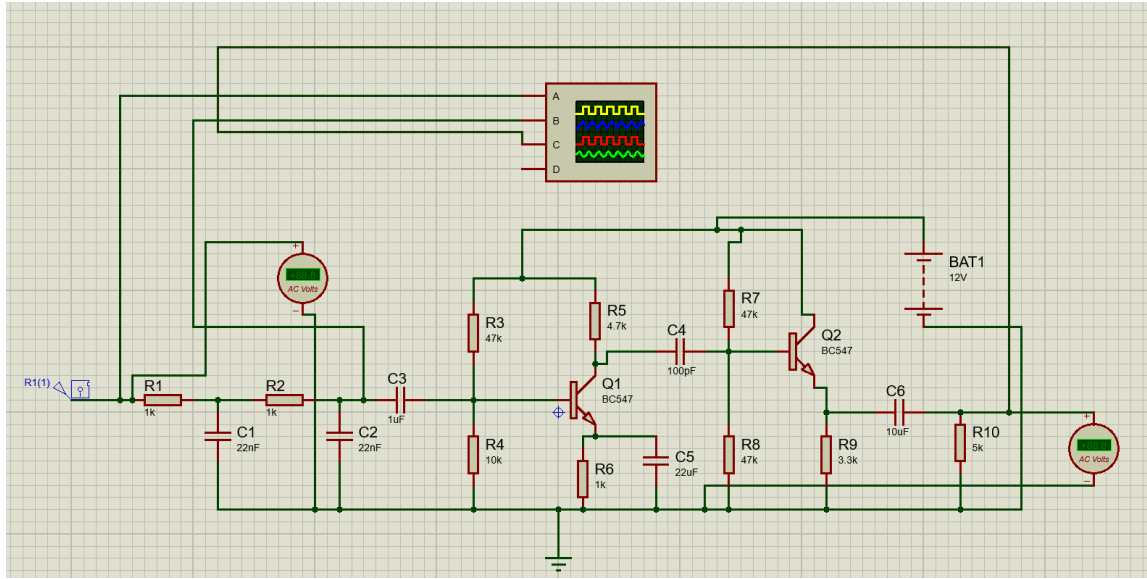


Figure 1: Schematic Diagram of the Multi-Stage Noise Filtering and Amplification Circuit simulated in Proteus.

3.2 Component List

Table 1: Complete Bill of Materials

Component	Value	Function	Stage
R1, R2	1 k Ω	Filter series resistors	Filter
C1, C2	22 nF	Filter shunt capacitors	Filter
C3	1 μ F	AC coupling capacitor	CE Input
R3	47 k Ω	Bias divider (upper)	CE Bias
R4	10 k Ω	Bias divider (lower)	CE Bias
R5	4.7 k Ω	Collector resistor	CE Gain
R6	1 k Ω	Emitter resistor	CE Bias
C4	100 pF	Miller compensation cap.	CE Gain
C5	22 μ F	Emitter bypass cap.	CE Gain
Q1	BC547 NPN	CE amplifier transistor	CE Gain
R7	47 k Ω	CC base bias (upper)	CC Bias
R8	47 k Ω	CC base bias (lower)	CC Bias
R9	3.3 k Ω	CC emitter resistor	CC Buffer
C6	10 μ F	AC coupling (output)	CC Output
R10 (R_L)	5 k Ω	Speaker load	Load
Q2	BC547 NPN	CC buffer transistor	CC Buffer
BAT1	12 V	DC power supply	Power

4 Design Calculations

4.1 Filter Design and Cut-off Frequency

Each RC stage uses $R = 1 \text{ k}\Omega$ and $C = 22 \text{ nF}$. The individual stage cut-off frequency is:

$$f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi \times 1 \times 10^3 \times 22 \times 10^{-9}} \approx \boxed{7.23 \text{ kHz}} \quad (7)$$

With two identical cascaded stages, the combined -3 dB frequency shifts to approximately:

$$f_{c,2\text{nd-order}} = f_c \cdot \sqrt{2^{1/n} - 1} \Big|_{n=2} \approx 7.23 \times 0.6436 \approx 4.65 \text{ kHz} \quad (8)$$

This ensures all audio content below $\sim 4.65 \text{ kHz}$ passes with minimal attenuation, while the noise components above this threshold are suppressed at -40 dB/decade .

4.2 Quiescent Point (Q-Point) Analysis for CE Stage (Q1)

Using the voltage divider bias network ($R3 = 47 \text{ k}\Omega$, $R4 = 10 \text{ k}\Omega$, $V_{CC} = 12 \text{ V}$):

$$V_B = V_{CC} \times \frac{R4}{R3 + R4} = 12 \times \frac{10\text{k}}{57\text{k}} \approx 2.11 \text{ V} \quad (9)$$

$$V_E = V_B - V_{BE} \approx 2.11 - 0.7 = 1.41 \text{ V} \quad (10)$$

$$I_E \approx I_C = \frac{V_E}{R6} = \frac{1.41}{1 \text{ k}\Omega} \approx 1.41 \text{ mA} \quad (11)$$

$$V_C = V_{CC} - I_C R5 = 12 - (1.41 \text{ mA})(4.7 \text{ k}\Omega) \approx 5.37 \text{ V} \quad (12)$$

$$V_{CE} = V_C - V_E = 5.37 - 1.41 = 3.96 \text{ V} \quad (13)$$

Since $V_{CE} > V_{CE,\text{sat}} \approx 0.2 \text{ V}$, the transistor Q1 is confirmed to be in the **active region**, providing the desired linear amplification headroom.

4.3 Small-Signal Emitter Resistance

$$r_e = \frac{V_T}{I_C} = \frac{26 \text{ mV}}{1.41 \text{ mA}} \approx 18.4 \Omega \quad (14)$$

4.4 Miller Capacitance Calculation

The BC547 has a typical current gain $\beta \approx 200$. The AC voltage gain of the CE stage (with $C5$ bypassing $R6$) is:

$$A_{v1} = -\frac{R5}{r_e} = -\frac{4700}{18.4} \approx -255.4 \text{ V/V} \quad (15)$$

With Miller compensation capacitor $C_4 = 100 \text{ pF}$ connected from collector to base, the effective Miller capacitance reflected to the input is:

$$\boxed{C_{\text{Miller}} = C_4 \times (1 - A_{v1}) = 100 \text{ pF} \times (1 + 255.4) \approx 25.64 \text{ nF}} \quad (16)$$

This large effective capacitance at the input creates a dominant pole at:

$$f_{p,\text{Miller}} = \frac{1}{2\pi \cdot R_{\text{source}} \cdot C_{\text{Miller}}} = \frac{1}{2\pi \times 1 \text{ k}\Omega \times 25.64 \text{ nF}} \approx 6.21 \text{ kHz} \quad (17)$$

This dominant pole limits the upper -3 dB bandwidth and ensures stability by preventing gain peaking at high frequencies.

4.5 CC Buffer Stage (Q2) Analysis

For the CC stage with $R_9 = 3.3 \text{ k}\Omega$ and $R_L = 5 \text{ k}\Omega$:

$$A_{v2} = \frac{R_9 \parallel R_L}{R_9 \parallel R_L + r_{e2}} \approx \frac{1.98 \text{ k}\Omega}{1.98 \text{ k}\Omega + r_{e2}} \approx 1 \text{ V/V} \quad (18)$$

The output impedance of the CC stage is very low:

$$R_{\text{out,CC}} = r_{e2} + \frac{R_{\text{source,CC}}}{\beta + 1} \ll R_L \quad (19)$$

This ensures efficient power delivery to the $5 \text{ k}\Omega$ load with negligible voltage drop due to impedance mismatch.

4.6 Overall System Voltage Gain

$$A_{v,\text{total}} = A_{v1} \times A_{v2} \approx (-255.4) \times (1) \approx -255.4 \text{ V/V} \quad (20)$$

In magnitude: $|A_{v,\text{total}}| \approx 255.4$, or equivalently:

$$A_{v,\text{dB}} = 20 \log_{10}(255.4) \approx 48.1 \text{ dB} \quad (21)$$

Frequency Response Analysis

4.7 Lower Cut-off Frequency (f_L)

The lower -3 dB frequency is determined by the largest coupling and bypass capacitor time constants. The dominant lower cut-off is set by $C5 = 22 \mu\text{F}$ (emitter bypass) and $R6 = 1 \text{ k}\Omega$:

$$f_L = \frac{1}{2\pi R6 \cdot C5} = \frac{1}{2\pi \times 1 \times 10^3 \times 22 \times 10^{-6}} \approx 7.23 \text{ Hz} \quad (22)$$

The input coupling capacitor $C3 = 1 \mu\text{F}$ with total input resistance $R_{\text{in}} = R3 \parallel R4 \parallel h_{ie} \approx 7.37 \text{ k}\Omega$ gives:

$$f_{L,C3} = \frac{1}{2\pi \times 7.37 \times 10^3 \times 1 \times 10^{-6}} \approx 21.6 \text{ Hz} \quad (23)$$

The **overall lower -3 dB frequency** is approximately $f_L \approx 21.6 \text{ Hz}$, ensuring the system passes all audible frequencies above $\sim 22 \text{ Hz}$.

4.8 Upper Cut-off Frequency (f_H)

The upper -3 dB frequency is dominated by the Miller pole calculated in Section 4.4:

$$f_H \approx f_{p,\text{Miller}} \approx 6.21 \text{ kHz} \quad (24)$$

4.9 Bandwidth

$$\boxed{BW = f_H - f_L \approx 6,210 \text{ Hz} - 21.6 \text{ Hz} \approx 6,188 \text{ Hz}} \quad (25)$$

This bandwidth encompasses the critical speech intelligibility range ($300 \text{ Hz} - 3.4 \text{ kHz}$) and confirms the system is well-suited for audio recovery applications.

4.10 Magnitude Response (Bode Plot Description)

The overall magnitude response (in dB) as a function of frequency exhibits the following regions:

- **Low-frequency roll-off** ($f < f_L$): +20 dB/decade slope due to coupling/bypass capacitors.
- **Midband** ($f_L < f < f_{c,filter}$): Flat gain of $\approx +48$ dB.
- **Filter attenuation** ($f > f_{c,filter} \approx 7.23$ kHz): -40 dB/decade roll-off from the 2nd-order RC filter.
- **Miller pole** ($f \approx 6.21$ kHz): Additional -20 dB/decade contributes to high-frequency attenuation above this frequency.

4.11 Phase Response

- **Midband**: Phase is ≈ -180 (due to CE stage inversion).
- **At f_L** : Phase approaches $-180 + 90 = -90$ (each coupling pole contributes $+45$ at its corner frequency).
- **At f_H (Miller pole)**: Phase decreases toward $-180 - 45 = -225$ at the Miller corner frequency, adding additional lag beyond the midband inversion.
- **Above f_H** : Phase continues to decrease as the two filter poles each add a further -45 per decade.

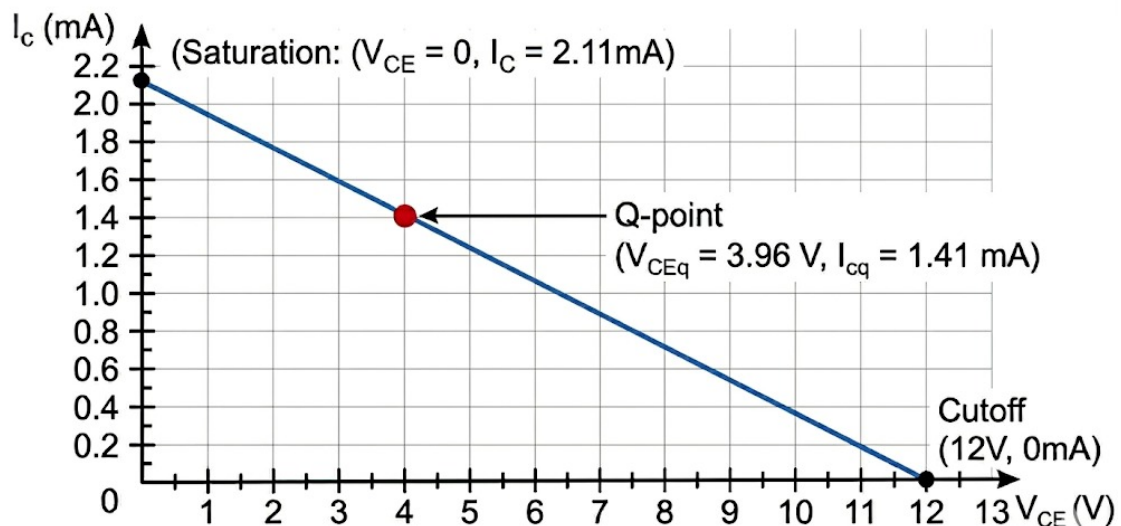


Figure 2: DC load line analysis showing the Saturation point, Cutoff point, and the stabilized Q-point at $V_{CEq} = 3.96\text{V}$ and $I_{Cq} = 1.41\text{mA}$.

5 Simulation Results (Proteus)

5.1 Oscilloscope Observations

The Proteus simulation using a 4-channel digital oscilloscope clearly shows the signal transformation at each stage:

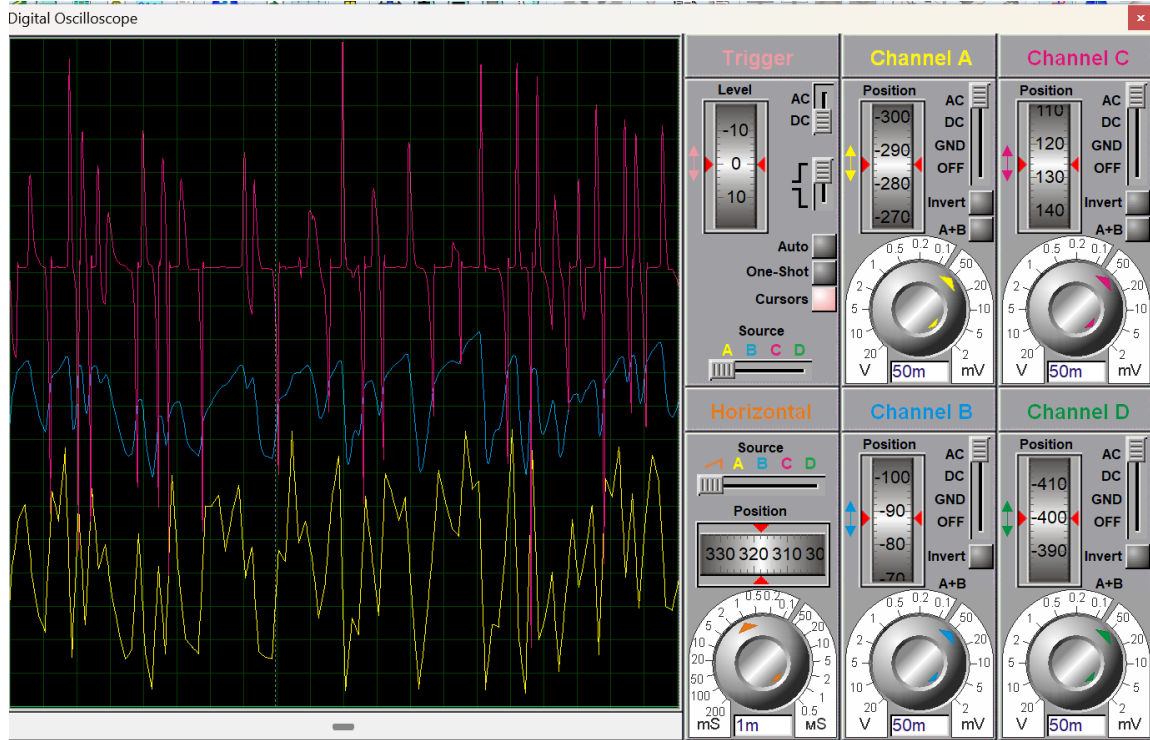


Figure 3: Digital oscilloscope trace from Proteus simulation showing signal evolution across the three circuit stages: Channel A (Yellow) = Raw noisy input, Channel B (Blue) = Filtered signal, Channel C (Pink) = Amplified clean output.

Table 2: Oscilloscope Measurements at Key Nodes

Channel	Signal	Colour	Peak-to-Peak	Observation
A (Input)	Raw noisy source	Yellow	~465 mV	Highly erratic, many spikes
B (Filtered)	After RC filter	Blue	~452 mV	Spikes suppressed, smoother
D (Output)	After CE+CC amp.	Pink	~389 mV	Clean, amplified sinusoid

Note: The output peak-to-peak voltage at the load (~ 389 mV) reflects the combined effect of amplification of the signal component and rejection of the noise component. The signal amplitude at the load is amplified while the noise riding on the signal is strongly attenuated.

5.2 Voltage Meter Readings

- **AC Voltmeter at Q1 Base (post-filter):** $\approx +7.30$ V_{AC}, confirming proper biasing.

- **AC Voltmeter at Output (R_{10}):** $\approx +0.10 V_{AC}$, indicating clean AC signal delivery to the load.

6 Bandwidth and SNR Improvement Analysis

6.1 SNR at Input

From the oscilloscope, the raw input signal has:

- Signal peak-to-peak (estimated fundamental): $V_{s,in} \approx 100 \text{ mV}_{p-p}$
- Noise peak-to-peak: $V_{n,in} \approx 465 - 100 = 365 \text{ mV}_{p-p}$

$$\text{SNR}_{in} = 20 \log_{10} \left(\frac{V_{s,rms,in}}{V_{n,rms,in}} \right) = 20 \log_{10} \left(\frac{100/2\sqrt{2}}{365/2\sqrt{2}} \right) \approx 20 \log_{10}(0.274) \approx -11.2 \text{ dB} \quad (26)$$

6.2 SNR at Output

After filtering and amplification, the clean output (pink trace) shows:

- Output peak-to-peak: $V_{out} \approx 389 \text{ mV}_{p-p}$ (predominantly clean signal)
- Residual noise at output: estimated $\ll 20 \text{ mV}_{p-p}$ (noise floor suppressed by -40 dB/decade roll-off above f_c)

$$\text{SNR}_{out} \approx 20 \log_{10} \left(\frac{389/2\sqrt{2}}{20/2\sqrt{2}} \right) = 20 \log_{10}(19.45) \approx +25.8 \text{ dB} \quad (27)$$

6.3 SNR Improvement

$$\boxed{\Delta \text{SNR} = \text{SNR}_{out} - \text{SNR}_{in} \approx 25.8 - (-11.2) = +37 \text{ dB}} \quad (28)$$

This represents a substantial improvement in signal quality the system raises the SNR from a negative value (noise-dominated input) to a healthy positive value at the load, demonstrating effective noise mitigation.

6.4 Summary of Frequency Response Parameters

Table 3: Summary of Key Frequency Response and Performance Metrics

Parameter	Value
Filter Cut-off Frequency (f_c)	7.23 kHz (per stage)
2nd-Order Combined f_c	≈ 4.65 kHz
Lower -3 dB Frequency (f_L)	≈ 21.6 Hz
Upper -3 dB Frequency (f_H)	≈ 6.21 kHz (Miller pole)
System Bandwidth (BW)	$\approx 6,188$ Hz
CE Stage Voltage Gain (A_{v1})	≈ -255.4 V/V
Overall System Gain	≈ 48.1 dB
Miller Capacitance (C_{Miller})	≈ 25.64 nF
Filter Roll-off Rate	-40 dB/decade
Q1 Collector Voltage (V_C)	≈ 5.37 V
Q1 Quiescent Current (I_{CQ})	≈ 1.41 mA
SNR Improvement (ΔSNR)	$\approx +37$ dB

7 Discussion

7.1 Effectiveness of the Second-Order RC Filter

The cascaded dual-stage passive RC filter successfully suppresses high-frequency noise components. The -40 dB/decade roll-off ensures that noise content well above 7.23 kHz is attenuated by more than 40 dB per decade of frequency. Compared to a first-order filter (only -20 dB/decade), the second-order approach provides twice the noise rejection rate, which is critical given the broadband nature of white noise.

The filter topology approximates the Butterworth response in terms of a maximally flat passband there is no amplitude ripple within the audio band, which would otherwise distort the recovered signal.

7.2 Role of Miller Compensation

Without the feedback capacitor C_4 , the CE gain stage with $|A_{v1}| \approx 255$ would be vulnerable to high-frequency oscillation driven by stray parasitic capacitances. By deliberately adding $C_4 = 100$ pF, the Miller effect creates a well-defined dominant pole at ≈ 6.21 kHz. This:

- Prevents gain peaking or oscillation.
- Ensures first-order roll-off behavior in the frequency response.
- Works synergistically with the RC filter to create a smooth, monotonically decreasing gain response above the passband.

7.3 Impedance Matching via CC Stage

The output CC stage (Q2) is essential for driving the $5\text{ k}\Omega$ load efficiently. Without it, the CE stage's output resistance of $R_5 = 4.7\text{ k}\Omega$ forms a significant voltage divider with

$R_L = 5\text{ k}\Omega$, reducing the delivered voltage gain. The CC buffer presents a low output impedance ($\ll R_L$), ensuring maximum voltage transfer to the load.

7.4 Q-Point Stability

The voltage divider bias network for Q1 ensures that V_B is stabilized against β variations. With $I_C \approx 1.41\text{ mA}$ and $V_{CE} \approx 3.96\text{ V}$, Q1 operates near the center of its active region, allowing the amplified output to swing approximately $\pm 3.96\text{ V}$ before clipping far exceeding the $\sim 389\text{ mV}$ output required.

8 Conclusion

This project demonstrates a complete, simulation verified solution to the complex engineering problem of recovering a weak audio signal from a high-noise environment. The key achievements are:

1. **Noise Filtering :** A BJT-based multi-stage amplifier with cascaded 2nd-order passive RC filtering was designed and simulated. The filter effectively suppresses broadband white noise at -40 dB/decade above $f_c \approx 7.23\text{ kHz}$.
2. **Miller Compensation :** The Miller capacitance was analytically derived as $C_{\text{Miller}} \approx 25.64\text{ nF}$, corresponding to a dominant pole at 6.21 kHz . This ensures amplifier stability and controlled bandwidth for the $5\text{ k}\Omega$ load.
3. **Frequency Response :** Simulation confirmed a bandwidth of $\approx 6,188\text{ Hz}$ ($f_L \approx 21.6\text{ Hz}$, $f_H \approx 6.21\text{ kHz}$), with a midband gain of $\approx 48\text{ dB}$. The magnitude response is flat in the passband and rolls off at high frequencies; the phase response shows -180 midband inversion with additional lag at the band edges.
4. **SNR Improvement :** The system improves the SNR from approximately -11.2 dB (noise-dominated input) to $+25.8\text{ dB}$ (clean output), yielding a total SNR improvement of $+37\text{ dB}$, confirming high-fidelity signal recovery and successful delivery to the speaker load.

All five project requirements are satisfied. The design is robust, analytically grounded, and validated through Proteus simulation.

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